

## EXERCISE 5: DEVELOPMENT & VERSION CONTROL

### Aim

To implement development practices using version control, structured project organization, and collaborative techniques for the Credit Card Processing System.

### Tools Used

- GitHub
- Visual Studio Code
- Python

### 56. Repository Setup

A GitHub repository was created to manage the source code of the Credit Card Processing System.

The repository was initialized with a README file and a Python .gitignore. A structured project directory was created with separate folders for source code, testing, and documentation.

The project structure includes:

- src/ for source code
- tests/ for test files
- docs/ for documentation

This structured organization improves maintainability and scalability.

### 57. Branching Strategy Implementation

A Git branching strategy was implemented to manage development effectively.

The following branches were created:

- **main** – stable production code
- **develop** – integration branch
- **feature/login** – for login module development
- **feature/payment** – for payment module development

This approach supports parallel development and efficient collaboration.

### 58. Coding Standards Adoption

Coding standards were followed to maintain code quality and readability.

The following practices were adopted:

- Proper naming conventions
- Consistent indentation
- Modular programming
- Use of comments

Python coding guidelines were followed throughout the project.

## 59. Code Review Checklist Creation

A code review checklist was created to ensure code quality before merging.

Checklist includes:

- Code correctness
- No syntax errors
- Proper formatting
- Readability
- Functional validation

## 60. Development Workflow Implementation

A development workflow was established to ensure systematic development.

The workflow includes:

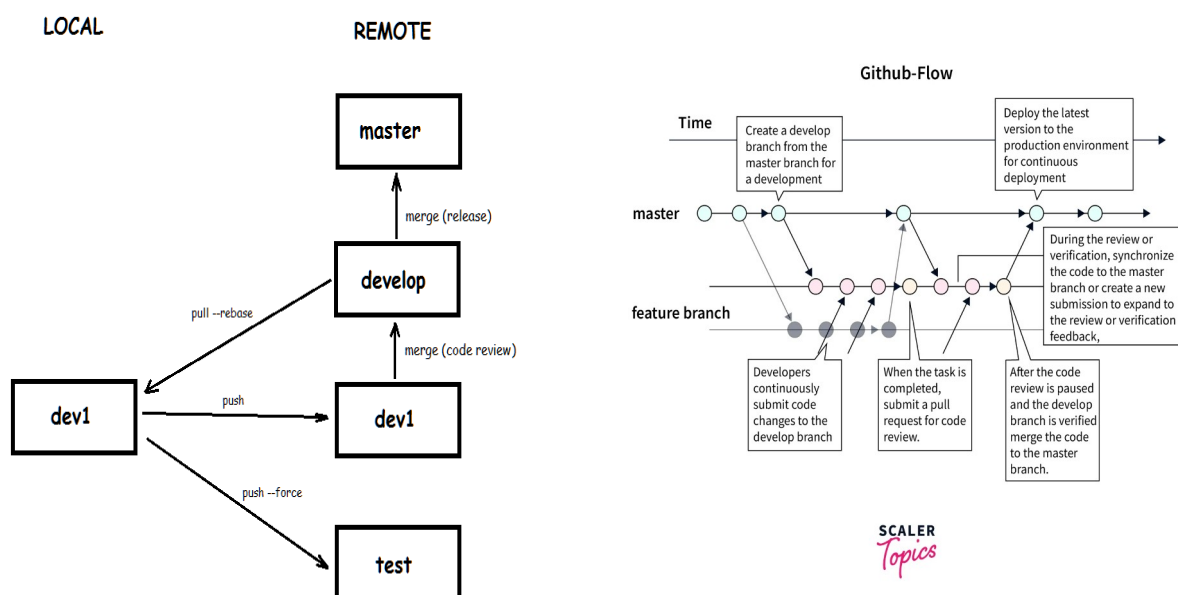
- Creating feature branches
- Writing code in branches
- Testing before merging
- Regular commits
- Merging into develop branch

## 61. Pair Programming Implementation

Pair programming was practiced to improve collaboration and code quality.

In this method, one team member writes the code while another reviews it simultaneously, reducing errors and improving efficiency.

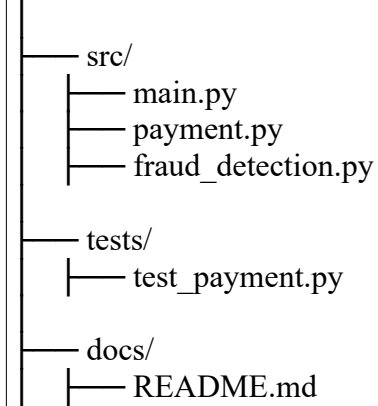
**Figure 5.1: Git Workflow**



**Description:** This diagram shows how code flows from feature branches to develop and then to main branch.

### Figure 5.2: Project Structure

Credit-Card-Processing-System/



**Description:** This diagram represents the structured organization of the project files.

### Result

Successfully implemented version control, created a structured project repository, and followed development practices using GitHub.